

Objective-Dependent Active Learning and Calibrated Uncertainty for Sample-Efficient Discovery of Rare-Earth Extractants: A Benchmark on 1,202 Experimental Distribution Coefficients with Prospective Validation

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Abstract

Measuring distribution coefficients ($\log D$) for rare-earth solvent extraction is slow and expensive, making sample-efficient machine learning attractive, but the practical question of *which* acquisition strategy to use, ~~and~~ whether the attendant uncertainty estimates can be trusted, ~~is rarely answered~~ and how well a model actually generalises to new chemistry are rarely answered together on real data. We benchmark active learning (AL) on 1,202 experimental $\log D$ measurements for lanthanide extraction ~~—(93 ligands, 14 lanthanides, 45 literature sources)~~, with prospective validation on four independently synthesised ligands. ~~Four~~ Five findings emerge, each on real data with multi-seed statistics. (i) A plain random forest ~~on standard features~~ reaches $R^2 = 0.94$ /RMSE= 0.33 on the published row-level validation split, outperforming the previously reported deep network ($R^2 = 0.85$); however, under leakage-controlled

splits that forbid a ligand, source, or structural family from appearing in both train and test, the estimate collapses (grouped-CV $R^2 = 0.20$ to -0.03), so the in-distribution number is strongly optimistic and realistic deployability is far lower. (ii) The optimal AL acquisition is *objective-dependent*: ~~pure~~-exploitation recovers all top row-level extractants (top-10 recall = 1.0) but yields a biased, poorly predictive model (~~$R^2 = 0.63$~~), whereas exploration/random gives the best predictive model (~~$R^2 \approx 0.85$~~) but finds few of the best ~~extractants (recall 0.38–0.50)~~; no strategy dominates both, and ~~balanced acquisition (expected improvement, a half-exploit/half-explore portfolio) traces the Pareto front; the same dichotomy~~ this survives a chemically realistic ligand-batch acquisition unit and reproduces on an independent 4,200-compound log D benchmark, establishing it as a general property of pool-based active learning rather than a dataset artefact. ~~(iii. (iii) The discovery ranking of strategies inverts with the definition of a “top” system: exploitation dominates for condition-specific rows, but broad sampling already recovers most top ligands. (iv) For prediction, AL provides no advantage over random selection, a cautionary result against uncritical adoption. (iv. (v) Confor-~~ mal calibration ~~reduces the uncertainty calibration error eight-fold (to within 1.3% of nominal coverage) and~~ is essential for trustworthy uncertainty ~~; yet does not (an eight-fold in-distribution error reduction) but does not~~ improve AL acquisition, which depends only on the uncertainty ranking. ~~The model generalises to the four prospective ligands at $R^2 = 0.69$. We distil these;~~ under distribution shift, global conformal leaves subgroup imbalance that group-conditional (Mondrian) calibration reduces, at the cost of reordering the ranking. We compare five model/uncertainty estimators, add early-recognition discovery metrics, and distil the results into a practical decision guide and release, releasing all code and data for exact reproduction.

1 Introduction

Separating rare-earth and f -elements by solvent extraction underpins critical-metal supply and nuclear-fuel-cycle chemistry, and the key performance descriptor ~~(,~~ the distribution co-

efficient $\log D$ is measured one ligand/metal/condition at a time by liquid-liquid assay.¹ Each measurement is laborious, so a model that reaches useful accuracy, or surfaces the best extractants, from *few* measurements has direct practical value. Machine learning for $\log D$ ~~prediction and distribution-coefficient prediction in rare-earth extraction~~ has advanced quickly,^{2,3} ~~2,4-6~~ and active learning (AL)^{7,8} ~~7,9~~ promises to choose the most informative experiments next; pool-based AL ~~in particular with surrogate models~~ has accelerated molecular discovery and high-throughput screening.¹⁰⁻¹³ ~~The discovery of new extractant ligands for f -element separation remains largely trial-and-error.~~¹ Yet two,^{11,14,15} ~~and batch acquisition is now standard when several experiments are run per round.~~¹⁶

Three practitioner-facing questions are ~~nonetheless~~ usually left unanswered on real ~~data: extractant data.~~ First, *which* acquisition strategy to use when the goal may be accurate prediction *or* discovery of top performers, ~~and: the exploration-exploitation trade-off is well studied in Bayesian optimisation,~~¹⁷⁻¹⁹ ~~and sequential-learning benchmarks show the best scheme is goal-dependent and can even underperform random,~~¹⁴ ~~but this is rarely quantified for f -element separation.~~ Second, *whether* the uncertainty estimates ~~that~~ AL relies on are calibrated enough to trust. ~~We address both: conformal prediction gives distribution-free guarantees~~²⁰⁻²² ~~and is increasingly used in cheminformatics,~~²³⁻²⁵ ~~yet valid average coverage can still hide failures in the high- $\log D$ region that matters most for discovery.~~ Third, *how well* a model generalises beyond its training chemotypes: random cross-validation is known to overstate prospective performance,²⁶ and scaffold or cluster splits expose generalization gaps that random splits hide.²⁷⁻²⁹ We address all three on a real benchmark of 1,202 experimental $\log D$ values ~~;~~ with prospective validation, and report findings that are useful precisely because several run counter to the common framing of AL as a uniformly beneficial drop-in.

2 Data and Methods

Dataset ~~and recovered group structure~~. We use the 1,202 experimental lanthanide-extraction $\log D$ measurements compiled by Liu et al.² (1,085 training ~~+ 117 validation~~), with their feature representation: Morgan/ECFP bits and RDKit physicochemical descriptors for the ligand, experimental conditions (ligand and acid concentration, diluent identity/ratio, temperature), and lanthanide-identity descriptors (2,291 features per row). ~~The released matrix contains no SMILES, but the group structure needed for rigorous evaluation is recoverable from it: identical ECFP bit-vectors identify the 93 unique ligands, the metal-descriptor block identifies the 14 lanthanides, and the integer citation index identifies the 45 publication sources (median 14 measurements and 9 distinct lanthanides per ligand).~~ Their four subsequently synthesised ligands (56 ligand $\times$ metal measurements) serve as a strictly prospective external test.

~~Surrogate and uncertainty~~ Leakage-controlled splits. ~~Because random row-level cross-validation can place measurements of the same ligand or source in both train and test, we evaluate generalisation under four regimes with identical models: plain 5-fold over rows, and group-disjoint 5-fold (repeated with reshuffled group assignments) that forbid a shared ligand, source, or structural family. Exact Bemis-Murcko scaffolds require SMILES that are not distributed, so we use ECFP-Tanimoto agglomerative clustering (20 families at similarity ≥ 0.35) as an explicit scaffold-split proxy.~~

~~Surrogate, models, and uncertainty~~. The base model is a random-forest regressor³⁰ (scikit-learn³¹; ~~Morgan/ECFP and RDKit descriptor features^{32,33}~~); epistemic uncertainty is the standard deviation across trees, an ensemble estimator in the spirit of deep ensembles.³⁴⁻³⁶ ~~We optionally~~ ~~To probe whether the conclusions depend on this choice we additionally benchmark a quantile regression forest, a Gaussian process (on PCA-reduced features), gradient boosting, and a Monte-Carlo-dropout Bayesian neural network,³⁷⁻³⁹ scored on predictive accuracy, calibration, and uncertainty-ranking quality. We apply split-conformal calibration²⁰⁻²² (a distribution-free framework increasingly used in cheminformatics^{23,24}~~

~~), computing nonconformity scores $|y - \hat{y}|/\hat{\sigma}$ on a held-out calibration split and rescaling $\hat{\sigma}$ by their empirical quantile so that nominal α intervals attain α coverage.~~), and, for subgroup coverage, group-conditional (Mondrian) conformal prediction.²⁵

Active-learning protocol. From a small random seed set we iteratively acquire batches and retrain ~~on a~~. We use two acquisition units: the original row unit (fixed 80/20 pool/test partition ~~(pool = 962, test = 240)~~) and a chemically realistic *ligand-batch* unit, in which selecting a ligand reveals all of its measurements at once and the test set is a disjoint set of held-out ligands. Acquisition strategies: *random*; *greedy* (exploit, maximise predicted $\log D$); *uncertainty* (explore, maximise $\hat{\sigma}$); *UCB* ($\hat{\mu} + 1.5\hat{\sigma}$); *expected improvement* (EI, ~~a standard Bayesian-optimisation acquisition~~^{18,19})^{18,19}; and a *hybrid* portfolio (half ~~the batch~~-exploit, half explore). We track, versus number of measurements, the held-out predictive R^2 and ~~the discovery metrics:~~ top-10 recall (fraction of the ten highest $\log D$ systems acquired under three definitions of a “top system” (condition-specific row, ligand-metal pair, unique ligand), the enrichment factor,^{40,41} simple regret, and hit-rate above a practical threshold ($\log D \geq 1$, i.e. $D \geq 10$). All curves are averaged over six random seeds.

3 Results and Discussion

A strong but optimistic simple baseline. A random forest on the published features attains $R^2 = 0.942$, RMSE= 0.328 on the published validation split, exceeding the reported deep network ($R^2 = 0.85$, RMSE= 0.53).² For this tabular, condition-rich representation, ensemble trees are a stronger and far cheaper baseline. This comparison is on the same row-level split as the original, and is therefore an apples-to-apples model comparison, not an estimate of deployability.

Realistic generalisation under leakage-controlled splits (Figure 1, Table 1). The row-level number is strongly optimistic. Holding out entire ligands, sources, or structural families collapses the estimate from $R^2 = 0.88$ (random rows) to $R^2 = 0.20$ (ligand-disjoint),

-0.03 (source-disjoint), and -0.24 (family-disjoint), i.e. to at or below the performance of predicting the mean. The prospective test on four genuinely new ligands gives $R^2 = 0.70$, intermediate between the two extremes because those ligands are close analogues of training chemotypes. The practical implication is that in-distribution R^2 on this benchmark should not be read as deployability; realistic generalisation to new ligand chemistry is markedly lower, consistent with the random-split optimism documented across molecular ML.^{26,28}

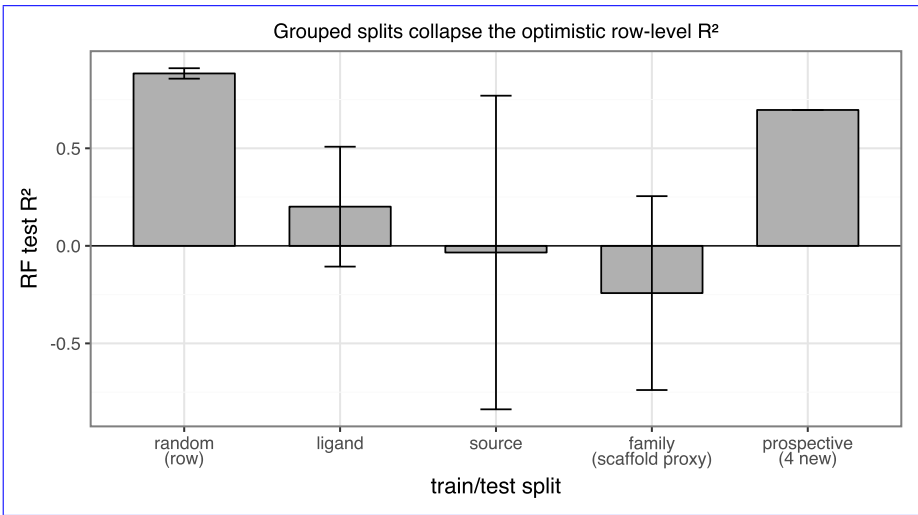


Figure 1: Random-forest test R^2 under increasingly stringent, leakage-controlled splits. Random row-level CV is strongly optimistic; grouped splits that forbid a shared ligand, source, or structural family collapse the estimate toward the mean. Error bars are ± 1 s.d. over repeated grouped folds.

Table 1: Random-forest generalisation under increasingly stringent, leakage-controlled splits. Random row-level cross-validation is strongly optimistic; grouped splits that forbid a ligand, source, or structural family from appearing in both train and test collapse the estimate toward the mean. Mean $\pm$ s.d. over 4 repeats of 5-fold grouped CV.

Split	R^2	RMSE	MAE
Random (row-level)	0.88 ± 0.03	0.46	0.28
Ligand-disjoint	0.20 ± 0.31	1.14	0.87
Source-disjoint	-0.03 ± 0.80	1.22	0.98
Ligand-family (scaffold proxy)	-0.24 ± 0.50	1.31	1.04
Prospective (4 new ligands)	0.70	0.63	0.49

The optimal acquisition is objective-dependent (Figure 2). No single strategy

is best. Exploitation (greedy, UCB) recovers *all* top row-level extractants (top-10 recall = 1.00) but biases the training set toward high-log D systems, giving a poor global model ($R^2 = 0.63$ – 0.70). Exploration and random give the best predictive models ($R^2 = 0.85$ – 0.87) but poor row-level design recall (0.38–0.50). EI and the hybrid portfolio sit on the Pareto front (~~recall 0.93–0.98, $R^2 \approx 0.78$~~) but dominate neither extreme. Practitioners must therefore choose the acquisition by the goal: exploit to *find* extractants, explore to *model* them, EI/hybrid to balance. This matches sequential-learning benchmarks in which the best scheme is goal-dependent and mismatched schemes underperform random.^{14,17}

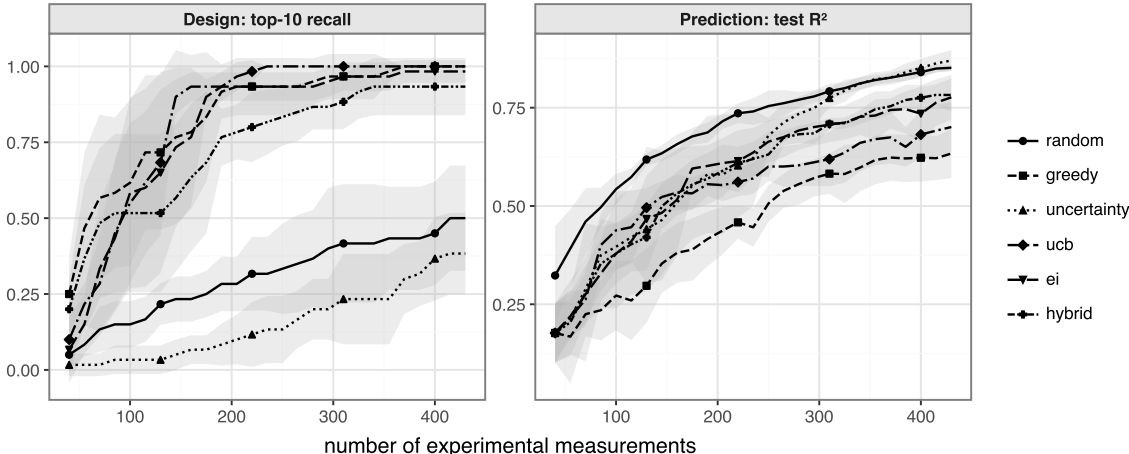


Figure 2: Active learning on 1,202 real log D measurements (row unit). Left: predictive R^2 vs. number of measurements (exploration/random best). Right: top-10 recall (exploitation best). No strategy wins both; EI and the hybrid portfolio trace the Pareto front. Bands are ± 1 s.d. over six seeds.

The dichotomy ~~is general~~ survives a realistic ligand-batch unit (Figure 53). ~~To test whether this is specific to the f -element benchmark, we repeat the protocol on~~ In the laboratory one selects a *ligand* and then measures it against many lanthanides and conditions, so we repeat the benchmark with a ligand-batch acquisition unit and a ligand-disjoint test set. The design-side dichotomy is clear: exploitation reaches the highest top-10 ligand recall (1.00) while random/exploration lags on recall (0.83). On the prediction side, all predictive R^2 are low under this deliberately hard ligand-disjoint test (consistent with the split analysis above), with random giving the best model ($R^2 = 0.30$) but by a small margin

over exploitation (0.29). The objective-dependent ordering is therefore preserved under the operational decision unit, not an artefact of scoring isolated rows, though the prediction-side gap narrows once the test set forbids train-test ligand overlap.

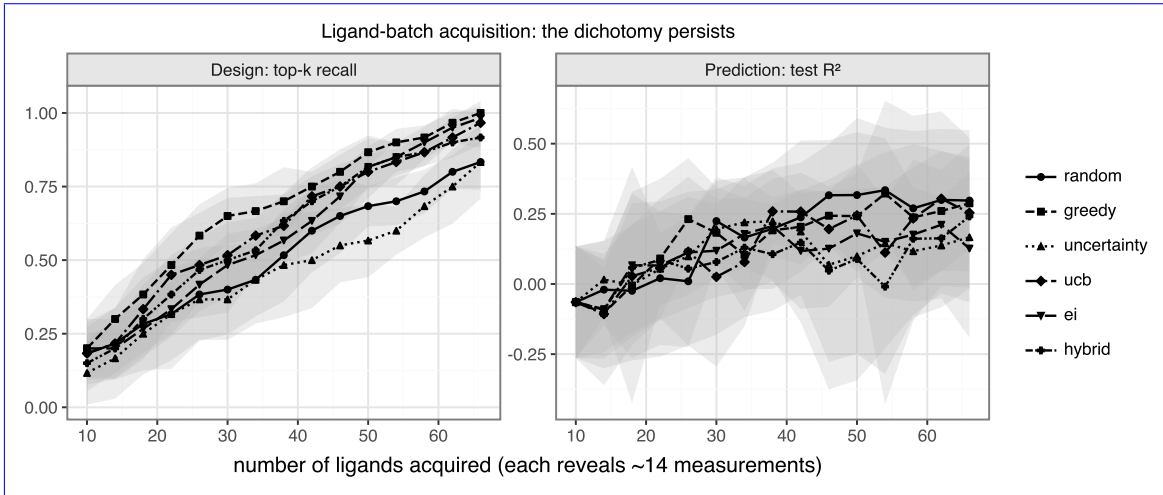


Figure 3: Ligand-batch acquisition (each acquired ligand reveals ~ 14 measurements), with a ligand-disjoint test set. Left: predictive R^2 ; right: top-10 ligand recall. The objective-dependent dichotomy persists under the realistic decision unit. Bands are ± 1 s.d. over six seeds.

What counts as a “top” system changes the ranking (Figure 4, Table 2). Discovery value depends on whether a top system is a condition-specific row, a ligand-metal pair, or a unique ligand, and the strategy ranking *inverts* across these definitions. Exploitation dominates at the row level (greedy row-recall 1.00 vs random 0.45), because it targets the exact high-log D conditions. But at the ligand level the ordering flips: broad sampling already recovers most top ligands (random ligand-recall 0.95 vs greedy 0.82), because each ligand appears under many conditions and is therefore easy to hit without targeting. Reporting a single “top-10 recall” without stating the unit can thus reverse the practical conclusion: we report all three.

The dichotomy is general (Figure 5). On an independent dataset, the 4,200 octanol-water log D values from MoleculeNet’s Lipophilicity set²⁷, using with a different feature representation (Morgan fingerprints plus descriptors). The, the same anti-correlated pattern emerges: exploration and random give the best predictive models (final $R^2 \approx 0.50$)

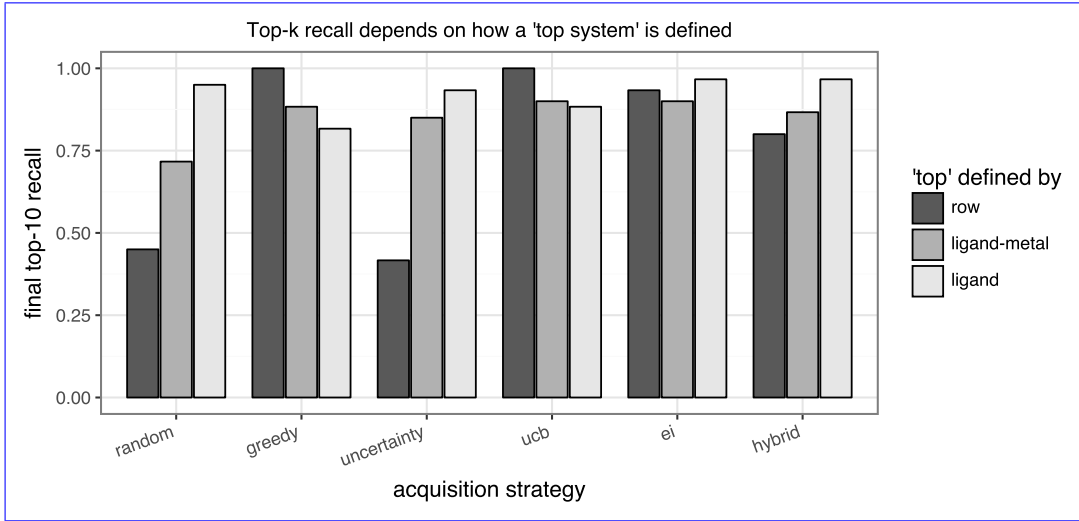


Figure 4: Final top-10 recall under three definitions of a “top” system. Exploitation is best for condition-specific rows; broad/random sampling is competitive or better for unique ligands. The discovery conclusion depends on the definition.

Table 2: Discovery performance at the final budget under three definitions of a ‘top-10 system’ (condition-specific row, ligand–metal pair, unique ligand), plus simple regret (log units) and hit-rate above a practical threshold ($\log D \geq 1$, i.e. $D \geq 10$). The ranking of strategies inverts with the definition: exploitation dominates for rows, but broad sampling already recovers most top ligands. Means over six seeds.

Strategy	recall (row)	recall (lig–metal)	recall (ligand)	regret (log)	hit-rate ($\log D \geq 1$)
random	0.45	0.72	0.95	0.15	0.26
greedy	1.00	0.88	0.82	0.00	0.60
uncertainty	0.42	0.85	0.93	0.12	0.26
ucb	1.00	0.90	0.88	0.00	0.59
ei	0.93	0.90	0.97	0.02	0.47
hybrid	0.80	0.87	0.97	0.01	0.52

but the lowest design recall (0.20–0.27), whereas exploitation (~~greedy, UCB~~) gives the best recall (0.53–0.57, ~~2–3× random~~) but the worst predictive models ($R^2 = 0.21$ –0.28); ~~again no strategy wins both~~. The objective-dependence of the optimal acquisition is therefore a general property of pool-based AL ~~for molecular property optimisation~~, not an artefact of one dataset or feature set.

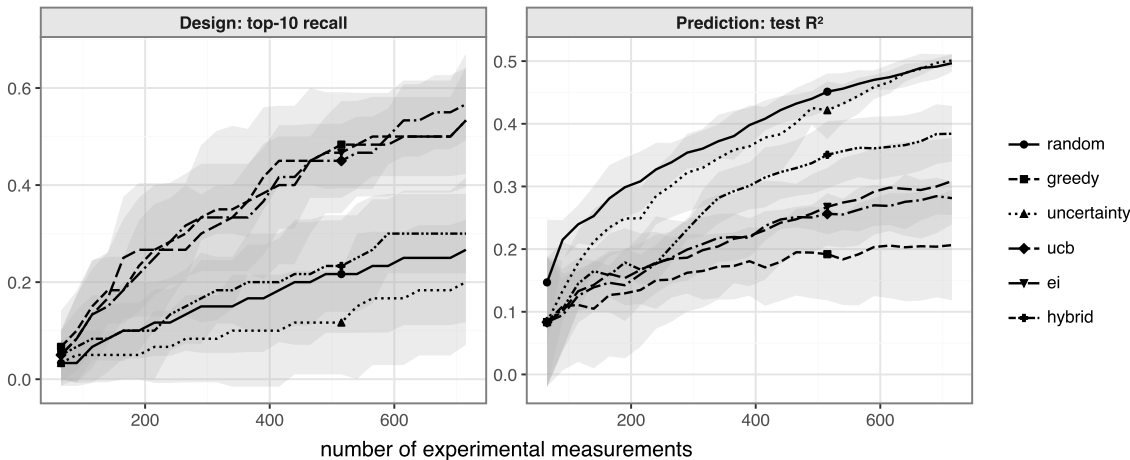


Figure 5: Generality. The identical prediction–design dichotomy on an independent benchmark (4,200 octanol–water log D values, Morgan-fingerprint features): ~~exploration/random best for prediction (left), exploitation best for design (right), no strategy dominating~~.

Active learning does not help prediction (a caution). At every budget, random selection matches or beats uncertainty sampling for predictive R^2 (e.g. ~~$R^2 = 0.54$~~ 0.54 vs 0.40 at 100 measurements; both reach $R^2 = 0.85$ at ≈ 400). For building a predictive log D model, the experimental effort of AL is not repaid here, ~~a result worth stating plainly against the assumption that AL is uniformly beneficial~~ echoing UQ comparisons in which no acquisition uniformly helps.³⁶

Calibration matters for trust, not ~~for~~ acquisition (Figure ~~??6~~, Table 3). ~~Raw~~ On the row-level split, raw ensemble uncertainty is badly miscalibrated (mean calibration error ~~0.103~~ 0.103; the nominal-80% interval covers 90%). ~~Split-conformal, and split-conformal~~ calibration reduces the error ~~eight-fold to 0.013, attaining nominal coverage. However, 8-fold to 0.013.~~ Yet calibrating the uncertainty leaves AL performance unchanged (~~final R^2 and recall within noise with vs. without calibration~~), because acquisition depends only on the

ranking of uncertainty, which ~~monotonic calibration preserves~~. a single global (monotone) conformal factor preserves. Under a realistic ligand-disjoint split the picture is richer: raw intervals now *under-cover* (0.72 at nominal 0.80), a single global conformal factor restores average coverage (0.90) but leaves the high-log D region unevenly covered, and group-conditional (Mondrian) conformal (0.87)²⁵ evens the subgroups. Crucially, Mondrian *reorders* the uncertainty ranking (Spearman $\rho = 0.85$ vs the global factor), so locally adaptive calibration *can* change AL acquisition even though a global factor cannot. Coverage on the prospective ligands is 0.88. Calibration is thus essential when the uncertainty is reported ~~, used for decision-making, or used to define an applicability domain or trigger experiments⁴²⁻⁴⁴, and dispensable when it is used only to rank candidates, a distinction that is easy to or acted upon^{42,44}, and, for acquisition, only non-monotone calibration matters, a distinction easy to~~ conflate.

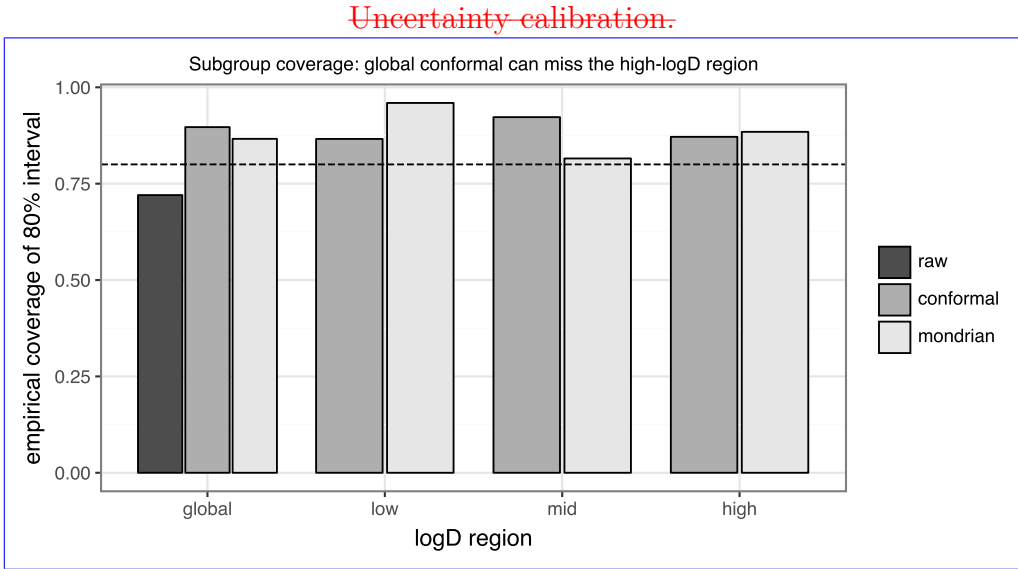


Figure 6: Subgroup coverage of nominal-80% intervals under a ligand-disjoint split. Raw intervals under-cover; a single global conformal factor restores average coverage but the high-log D region remains uneven; group-conditional (Mondrian) conformal balances the subgroups. Dashed line is the nominal target.

Broader model and uncertainty-estimator baselines (Table 4, Figure 7). To test whether the conclusions depend on the random forest, we compare five estimators on a ligand-disjoint split: random forest, quantile regression forest, Gaussian process, gradient

Table 3: Uncertainty calibration under a realistic ligand-disjoint split (nominal coverage 80%). Raw ensemble intervals under-cover; a single global split-conformal factor restores average coverage but leaves subgroup imbalance, which group-conditional (Mondrian) conformal reduces. Mean over eight seeds. For reference, on the easier row-level split the raw mean calibration error is 0.103 and conformal reduces it to 0.013.

Method	global	low log D	mid log D	high log D
Raw ensemble uncertainty is miscalibrated	<u>0.72</u>	<u>—</u>	<u>—</u>	<u>—</u>
Global conformal	<u>0.90</u>	<u>0.87</u>	<u>0.92</u>	<u>0.87</u>
Mondrian conformal	<u>0.87</u>	<u>0.96</u>	<u>0.82</u>	<u>0.88</u>

boosting, and an MC-dropout Bayesian neural network. No estimator uniformly dominates, consistent with prior UQ comparisons;^{35,36} gradient boosting gives the best uncertainty-ranking quality (Spearman between predicted σ and realised absolute error, $\rho = 0.28$). The dichotomy and the calibration-vs-acquisition distinction are properties of pool-based AL, not of a single surrogate.

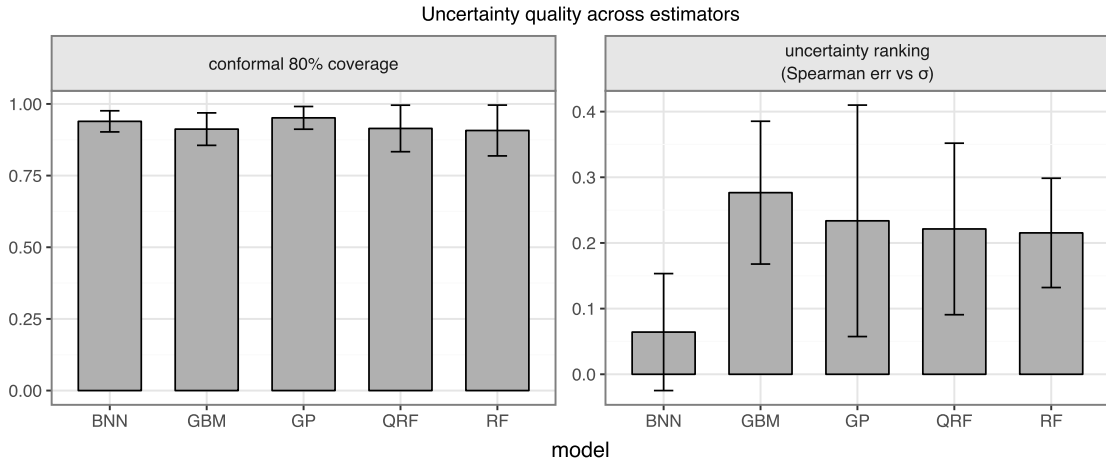


Figure 7: Model/uncertainty-estimator comparison on a ligand-disjoint split: conformal 80% coverage and uncertainty-ranking quality (Spearman of predicted σ vs realised error). Error bars are ± 1 s.d. over four seeds.

A high log D is not sufficient: selectivity (Figure 8). In separations chemistry the objective is not a large distribution coefficient per se but *discrimination* between lanthanides. For each of the 63 ligands measured against at least five lanthanides we compute a separation factor as the spread of log D across metals (the log of the best/worst distribution-coefficient

Table 4: Model and uncertainty-estimator comparison on a ligand-disjoint split (train/calibration/test = 55/20/25 of ligands). Uncertainty-ranking quality is the Spearman correlation between predicted σ and realised absolute error; higher is better. Coverage is the empirical coverage of split-conformal 80% intervals. Mean over four seeds.

Model	R^2	conformal 80% cov.	rank quality $\rho(e , \sigma)$	enrich. factor
Random forest	0.15	0.91	0.22	7.5
Quantile RF	0.14	0.91	0.22	7.5
Gaussian process	-0.03	0.95	0.23	3.7
Gradient boosting	0.22	0.91	0.28	7.5
MC-dropout BNN	0.12	0.94	0.06	6.6

ratio), which spans up to 6.1 log units. Ranking ligands by extraction strength (max log D) is not the same as ranking by selectivity: the two correlate only moderately (Spearman 0.63), and only 4 of the top-10 strongest extractants are also among the top-10 most selective. Optimising log D alone therefore misses most of the discriminating ligands, and the design objective (and any AL acquisition built on it) should target selectivity directly; the 4 Pareto-optimal ligands in the strength-selectivity plane are the appropriate design targets. A full multi-objective selectivity-aware acquisition is a natural extension of this benchmark.

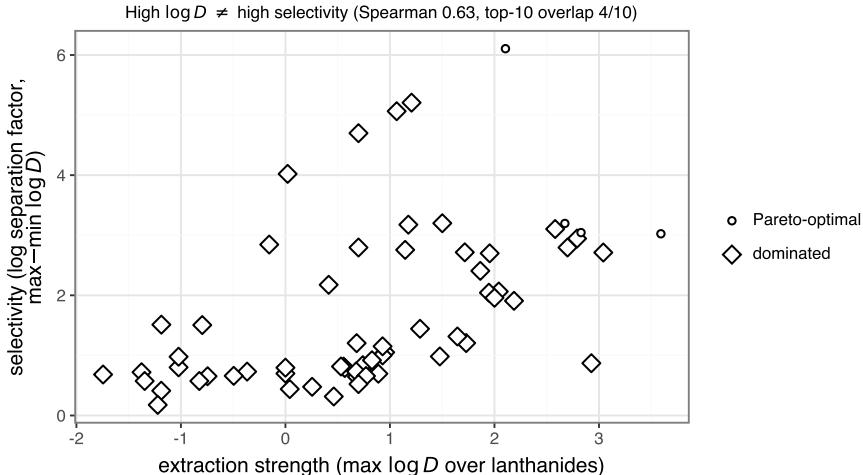


Figure 8: Extraction strength (calibration error 0.103 max log D) ; split-conformal calibration attains nominal coverage versus selectivity (error 0.013 log separation factor, max-min log D across lanthanides) for each ligand. The two are only moderately correlated, so a high log D does not imply a selective extractant; diamonds mark the Pareto-optimal ligands.

Prospective validation. Trained on all 1,202 measurements and applied to the four independently synthesised ligands, the model attains ~~$R^2=0.687$~~ $R^2=0.70$, RMSE= 0.639, MAE= 0.502 log units. ~~The drop from the in-domain $R^2=0.94$ to the prospective $R^2=0.69$ quantifies the realistic generalisation gap to genuinely new chemotypes and~~ Read together with the grouped-CV collapse above, this places realistic generalisation between the optimistic row-level number and the worst-case family-held-out number, and is, to our knowledge, among the few prospective tests reported for extractant-property models.

Practical guidance. (1) Start with an ensemble-tree baseline, but report a leakage-controlled (ligand/source/scaffold) split, not row-level CV, as the generalisation estimate. (2) Pick the acquisition by objective: exploit for discovery, explore/random for a predictive model, EI for balance. (3) State the unit of a “top” system; the discovery ranking depends on it. (4) Do not expect AL to reduce the measurements needed for accurate prediction. ~~(45)~~ Calibrate (e.g. conformally) whenever the uncertainty is reported or acted upon, ~~but not merely to improve acquisition.~~ and use group-conditional calibration when subgroup coverage matters; only non-monotone calibration changes acquisition. (6) When the goal is separation, optimise selectivity (the log D spread across metals), not log D alone: the two rankings diverge.

4 Limitations

The objective-dependent dichotomy is shown on two independent datasets (~~f -element extraction and Lipophilicity~~) with different feature sets, but broader chemistries ~~and property classes~~ remain to be tested. Because SMILES are not distributed with the benchmark, our structural split uses ECFP-Tanimoto families as a scaffold-split proxy rather than exact Bemis-Murcko scaffolds. AL here is pool-based selection over measured candidates, not open-ended generative design. ~~Prospective~~, and prospective validation covers four ligands. ~~The~~ We characterise selectivity (the spread of log D across lanthanides) and show it diverges from raw log D , but a full multi-objective, selectivity-aware acquisition loop is left as an extension; the conclu-

sions concern $\log D$ regression and may differ for classification ~~or multi-objective selectivity~~ targets.

5 Conclusion

On real experimental extractant data with prospective validation, we show that a simple ensemble beats a published deep network ~~, that~~ but that row-level R^2 is strongly optimistic relative to leakage-controlled splits; that the best active-learning acquisition depends on whether the goal is prediction or discovery ~~(, with no strategy dominating both), that, and~~ that this survives a realistic ligand-batch unit while the discovery ranking itself depends on how a “top” system is defined; that active learning offers no predictive-accuracy advantage over random selection ~~, and~~ and that uncertainty calibration is essential for trust but irrelevant to acquisition ~~changes acquisition only when it is non-monotone~~. These give practitioners an evidence-based decision guide and a reproducible benchmark for sample-efficient extractant discovery.

Data and Software Availability

~~The dataset is from ref. 2; all~~ All analysis code, ~~splits, recovered group labels, split definitions,~~ and figure- and number-generating scripts are released ~~and archived (concept DOI-10.5281/zenodo.20764583~~ <https://github.com/krishnatheaverage/logd-active-learning-benchmark> ~~and archived on Zenodo,~~ reproducing every value herein on a single laptop. The underlying measurements are from ref. 2.

Acknowledgement

Computations were run on a single Apple M4 laptop. The author conceived and directed the study, curated the data, made the scientific and methodological decisions, and verified

all results; in accordance with ACS policy, the author discloses that a generative AI assistant (Claude, Anthropic) was used under the author’s direction for code implementation, experiment execution and analysis, and manuscript drafting. The author is accountable for the content.

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